

AXON Network Tokenomic Framework: A Proof-of-Learning Model for a Decentralized AI Value Network

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Abstract

With the rapid development of Artificial Intelligence (AI), the centralization of its underlying infrastructure has become increasingly pronounced, giving rise to multiple challenges, including data monopolies, computational barriers, inequitable value distribution, and privacy violations. To address these issues, this paper proposes AXON Network, a decentralized AI value network aimed at fostering a contribution-oriented, fair, and transparent ecosystem. The core of AXON network lies in its token economic model, where the \$AXON token serves as the cornerstone for network security, governance, and application-layer utility. We introduce the novel Federated Attribution Proof (FAP) algorithm, which integrates the privacy-preserving features of federated learning with a precise contribution measurement mechanism. This enables the accurate quantification and attribution of the effectiveness of contributions from data, algorithms, and computing power, measured by their impact on model training. This paper details the design of core system components—the Domain-Library, Data-Feed, and Compute-Grid—to construct a multi-layered value capture and distribution framework. By employing an attribution mechanism based on semantic fingerprinting and gradient tracking, alongside a dynamically balanced tokenomics model, AXON provides a robust theoretical foundation and a viable practical framework for the sustainable development and large-scale collaboration of decentralized AI networks. It aims to democratize the creative potential of AI, liberating it from the control of a few tech giants to build a more open, inclusive, and efficient intelligent future.

1 Introduction

1.1 Background

Artificial intelligence has become the core engine of the latest technological revolution and industrial transformation. However, the current AI development paradigm is highly dependent on centralized entities, leading to a series of systemic problems. First, data monopolies and information silos have become increasingly severe. High-quality, domain-specific proprietary data is controlled by a handful of technology giants. Concurrently, the immense value embedded in the vast data generated by individuals in the digital realm is exploited by platforms without fair compensation, leaving users not only unrewarded but also exposed to risks of continuous privacy monitoring.

Second, the over-concentration of AI computing power constitutes another significant barrier. The training and inference of deep learning models demand enormous computational resources, with their growth in demand outpacing Moore’s Law. This has largely rendered AI innovation the exclusive domain of capital-intensive giants. Consequently, small and medium-sized enterprises, research institutions, and individual developers are often unable to afford the prohibitive costs of computing power, which stifles innovation.

Finally, existing blockchain consensus mechanisms are misaligned with the needs of AI development. Consensus mechanisms like Proof-of-Work (PoW), while ensuring network security, consume vast amounts of

electricity on hash computations that lack practical application value, resulting in significant energy waste and environmental concerns. While Proof-of-Stake (PoS) mechanisms improve energy efficiency, their inherent “rich-get-richer” dynamic can exacerbate wealth concentration. This may lead to the centralization of network governance, contradicting the foundational ethos of decentralization.

1.2 Motivation

In response to these challenges, the core motivation behind AXON Network is to design an economic system that deeply integrates the blockchain consensus process with AI value creation. Our central philosophy is that achieving consensus should be a value-creative process, not merely an act of resource consumption. To this end, we propose a generalized “Proof-of-Learning” (PoL) mechanism. Its essence is to transform the non-specific computational tasks of traditional PoW into beneficial training and validation tasks for AI models, a concept that aligns with the principle of “Proof of Useful Work.” Through this approach, the computing power that secures the network simultaneously contributes “intelligence” to the entire ecosystem, achieving a fundamental improvement in resource-use efficiency.

Furthermore, to address the issue of inequitable value distribution, we are committed to creating a network that can precisely track and reward all valuable contributions. Whether from an institution providing high-quality datasets, a node contributing idle computing power, or an individual user offering implicit knowledge through application interactions, all contributions (defined by their effectiveness for model training) must be quantified and fairly compensated. This necessitates the design of an innovative algorithm that enables accurate attribution within a robust privacy-preserving framework, ensuring that value distribution is transparent, just, and efficient.

1.3 Major Contributions

To realize this vision, the primary theoretical and technical contributions of this paper are fourfold:

1. **Proposal of the Federated Attribution Proof (FAP) Algorithm:** We designed an innovative attribution algorithm based on the federated learning framework. Without exposing raw user data, it accurately quantifies the marginal contribution of different data sources to model performance via gradient analysis and semantic fingerprinting, effectively resolving the fundamental tension between collaboration and privacy while establishing a robust defense against data duplication attacks.
2. **Design of \$AXON Token Economic Model:** We constructed a token system where the \$AXON token functions as the universal medium for network security, governance staking, core contribution incentives, and as the native asset for paying network transaction fees.. This unified model ensures that all value accrues directly to \$AXON.
3. **Construction of a Three-Layer Value Capture System:** We have clearly defined three core contribution layers: computing power, knowledge vaults, and personal data pods. By designing specialized quantification and incentive mechanisms for each layer, the system can comprehensively capture and integrate diverse forms of value—from foundational hardware resources to high-level knowledge applications—thereby forming a self-reinforcing value creation loop.
4. **Innovation in Dynamic Economic Regulation and Sustainability:** We have introduced a dynamic issuance model and a veToken governance model. These adaptive economic regulators collectively establish a long-term, robust economic system that ensures the network’s sustainable development.

2 Related Work and Problem Analysis

2.1 Economic Deficiencies of Existing Consensus Mechanisms

2.1.1 The Energy Efficiency Problem of Proof-of-Work (PoW)

PoW ensures ledger immutability through hashrate competition. However, as previously noted, its immense energy consumption is a widely recognized drawback. This model directs vast computing power toward solving mathematical puzzles with no intrinsic application value. The marginal security gained is far outweighed by the opportunity cost—namely, that this same computing power could be used for tasks like AI training, which would generate significant socioeconomic benefits.

2.1.2 The Centralization Trend of Proof-of-Stake (PoS)

PoS replaces computing power with token holdings as the determinant for consensus, significantly reducing energy consumption. However, it introduces a “Matthew effect” for capital. Validators’ rewards are proportional to their stake, creating a positive feedback loop where a few “whale” nodes can eventually dominate the network, undermining its decentralization and censorship resistance.

2.2 Incentive Mismatch in AI Computing Power Projects

In recent years, several projects aiming to build decentralized AI or computing power markets have emerged. While they have partially mitigated the concentration of computing power, they generally face fundamental challenges with their incentive mechanisms.

- **Difficulty in Quantifying Contribution:** Most existing projects reward participants based on simple metrics like task completion or resource uptime, making it difficult to accurately assess the “quality” or “effectiveness” of a contribution. For instance, in many decentralized intelligence networks, effectively evaluating the relative “intelligence” or training effectiveness of one AI model compared to others and rewarding it accordingly remains a core challenge.
- **The Inherent Conflict Between Privacy and Collaboration:** The advancement of AI depends on large-scale collaborative data training, which is in sharp conflict with the need to protect user privacy. Although technologies like federated learning offer a potential solution, the challenge lies in creating an incentive structure that motivates collaboration on sensitive data. This is only possible when robust, non-negotiable privacy guarantees are built into the protocol’s core, ensuring that mutually distrusting parties in an open, decentralized network can contribute safely and have the effectiveness of their contributions distributed fairly.
- **Lack of Transparency in Value Distribution:** Within the complex AI model training pipeline, a contributor often cannot trace the ultimate impact on model performance generated by their contribution. This disconnect in the value chain severely diminishes the motivation for participation.
- **Data Sybil Attacks:** A contributor might submit the same dataset multiple times through different nodes or accounts to fraudulently claim multiple rewards. This undermines the fairness of the reward system and dilutes the value of genuine, unique contributions. A robust system must be able to detect and prevent such behavior at the protocol level.

2.3 Technical Challenges in Data Valorization

Data is often hailed as “the oil of the new era”, yet quantifying its effectiveness is far more complex than for physical commodities. Quantifying abstract dimensions—such as data quality, scarcity, and relevance to a specific training task—into a computable, attributable “Effectiveness Index” presents enormous technical challenges. The privacy dimension is particularly critical, as the effectiveness of data is often proportional to its sensitivity. Developing technical means, such as Differential Privacy, to provide quantifiable protection for privacy, thereby unlocking data’s training potential for secure computation and fair incentives, is key to building a trustworthy data market. The design of AXON Network aims to systematically address these challenges by establishing a unified, transparent, and fair market for the quantification and attribution of the training effectiveness of data, computing power, and AI models.

3 Core Concept Definitions

To establish the economic and technical framework of AXON Network, we must first formally define the system’s core participants and effectiveness-bearing entities. This chapter details three fundamental components: the Domain-Library, the Data-Feed, and the Compute-Grid. Together, these form the infrastructure for the flow of effectiveness across data, knowledge, and computation within the AXON network.

3.1 Domain-Library

Definition 3.1: A **Domain-Library** is a domain-specific, structurally encapsulated, on-chain knowledge container. It is not merely an aggregate of high-quality datasets but a composite digital asset embodying domain models, metadata, contributor stakes, and the state of federated learning tasks. Its core design objective is to enable the verifiability, traceability, and attribution of the training effectiveness of specialized knowledge.

The design of the Domain-Library addresses the fundamental problem in AI development where high-quality, domain-specific data is both scarce and fragmented. By encapsulating data, models, and contribution records as standardized on-chain objects, it provides the foundational building blocks for decentralized AI collaboration. Each Domain-Library represents a repository of knowledge for a specific domain, its effectiveness collectively affirmed by the network through a rigorous evaluation mechanism, thereby serving as a source of training effectiveness within the \$AXON ecosystem.

A Domain-Library, DL_i , is mathematically defined as an octuple:

$$DL_i = \{I_i, D_i, M_i, E_i, C_i, T_i, S_i, F_i\} \quad (1)$$

Where:

- DL_i : Represents the i-th Domain-Library.
- I_i : A globally unique identifier generated from a content hash (e.g., an IPFS CID), ensuring its immutability and verifiability.
- D_i : The collection of knowledge datasets, with raw data stored via a distributed storage protocol.
- M_i : Metadata describing the Domain-Library, including its domain, tags, version, and usage license.
- E_i : An Effectiveness Index assessed by the verifier network, serving as the core parameter for assessing the vault’s contribution..

- C_i : The list of contributors, recording all addresses (a_j) that participated in the vault's creation and their corresponding contribution weights (w_j).
- T_i : A timestamped sequence of key operations, providing a historical record of its creation, updates, and validations.
- S_i : The staking status of the Domain-Library, indicating the quantity of \$AXON tokens providing it with economic security.
- F_i : The federated learning parameter set, storing specific parameters for learning tasks associated with the vault.

The comprehensive Effectiveness Index of a Domain-Library, E_i , is determined by a multi-stage, feedback-driven evaluation framework that combines an initial static assessment, post-use dynamic attribution, and community consensus verification. Its evaluation function is:

$$E_i = \alpha \cdot E_{i,\text{prior}} + \beta \cdot E_{i,\text{post}} + \gamma \cdot E_{i,\text{peer}} \quad (2)$$

Where:

- E_i : Represents the final comprehensive Effectiveness Index of Domain-Library V_i .
- $E_{i,\text{prior}}$: The a priori Effectiveness Index, representing the initial evaluation based on the vault's intrinsic attributes at the time of submission.
- $E_{i,\text{post}}$: The a posteriori Effectiveness Index, a dynamic feedback score derived from the measurable effectiveness the knowledge demonstrates in real-world AI applications.
- $E_{i,\text{peer}}$: The peer-review score of effectiveness, an assessment conducted by network verifiers or domain experts, thereby incorporating human expertise.
- α, β, γ : Weighting coefficients for each score, satisfying $\alpha + \beta + \gamma = 1$, which can be dynamically adjusted by the \$AXON DAO through governance.

The a priori Effectiveness Index, $E_{i,\text{prior}}$, is calculated based on the static characteristics of the data itself:

$$E_{i,\text{prior}} = \omega_1 \cdot A_i + \omega_2 \cdot C_i + \omega_3 \cdot U_i + \omega_4 \cdot V_i \quad (3)$$

Where:

- A_i, C_i, U_i, V_i : Respectively represent the scores for the Domain-Library's Accuracy, Completeness, Uniqueness, and Verifiability.
- $\omega_1, \omega_2, \omega_3, \omega_4$: The weighting coefficients for each of these static metrics.

The a posteriori Effectiveness Index, $E_{i,\text{post}}$, is centered on the Federated Attribution Proof (FAP) mechanism (detailed in Chapter 4), which quantifies the actual effectiveness of a Domain-Library for each AI service output on the network:

$$E_{i,\text{post}} = \frac{\sum_{t=1}^T F(o_t, DL_i) \cdot I(o_t)}{\sum_{t=1}^T F_{\text{total}}(o_t)} \quad (4)$$

Where:

- $F(o_t, DL_i)$: The attributed effectiveness of Domain-Library DL_i to a given AI output o_t , as calculated by the attribution algorithm.
- $I(o_t)$: The significance or weight (Impact) of the output o_t .
- $F_{\text{total}}(o_t)$: The total attributed effectiveness from all Domain-Libraries in the network to the output o_t .
- This formula calculates the weighted average effectiveness rate of V_i over a time period T .

This tripartite evaluation model ensures that the assessment of knowledge effectiveness is both comprehensive and fair. It effectively prevents ineffective data from being over-rewarded while accurately identifying and rewarding highly effective knowledge that demonstrates significant impact in practical applications.

3.1.1 Anti-Duplication Mechanism

To protect the integrity of the contribution ecosystem and prevent data Sybil attacks, AXON implements a two-layer anti-duplication mechanism during the Domain-Library validation process:

1. **Layer 1: Content Hash Verification.** The protocol maintains an on-chain registry of all submitted Domain-Library content hashes (I_i). When a new vault is proposed, its hash is checked against this registry. If an exact match is found, the submission is automatically rejected as a duplicate. This provides an absolute defense against identical data being submitted multiple times.
2. **Layer 2: Semantic Uniqueness Assessment.** For submissions that pass the hash check (i.e., are not exact duplicates), the network performs a more sophisticated semantic analysis. It calculates a semantic fingerprint (as detailed in Sec 4.3.1) of the new dataset and compares it against the fingerprints of existing, validated Domain-Libraries. The result of this comparison directly informs the 'Uniqueness' score (U_i) in the $E_{i,\text{prior}}$ calculation. A high degree of semantic similarity to an existing vault will result in a significantly lower U_i score, drastically reducing the potential rewards and disincentivizing attempts to submit slightly altered versions of existing data.

To perform this cross-library similarity check without compromising the confidentiality of the underlying data, the protocol employs a combination of advanced cryptographic and economic mechanisms. The core of this is a Privacy-Preserving Set Intersection (PSI) protocol, which allows the network to assess similarity without revealing the raw data fingerprints of the respective libraries. The specific implementation includes:

- A differentially private estimation of the Jaccard similarity index, potentially using Bloom Filter-based PSI protocols.
- A punitive scoring function that sharply reduces the Uniqueness score (U_i) as similarity increases. For instance, for a similarity score $J > 0.7$, the uniqueness score may be scaled by a factor of $(1 - J)^2$.
- An "Originality Bond" that creators must stake when submitting a new Domain-Library. This bond serves as collateral and can be slashed if the submission is successfully challenged and proven to be a malicious duplicate by the community.
- A "Near-Duplicate Arbitration" mechanism, where a randomly selected committee of verifiers can be called upon to review statistical samples and resolve disputes over alleged near-duplicates.

A formal analysis of the cryptographic protocols and their privacy parameters will be detailed in a dedicated AXON AI Documentation.

This dual-layer system ensures that rewards are directed towards genuinely new and unique contributions, safeguarding the economic value of all participants' efforts.

3.1.2 Assetization via Knowledge NFTs (LNFTs)

To enhance the asset-like properties and composability of Domain-Libraries, each vault is represented on-chain as a unique Knowledge NFT (LNFT). The LNFT acts as a deed of ownership, encapsulating the vault’s core metadata (I_i, M_i), its latest validated Effectiveness Index (E_i), and the access rights to its underlying data. This design innovation transforms a static data container into a dynamic, liquid digital asset. LNFTs can be traded on open markets, used as collateral in DeFi protocols, or leased to other users, allowing owners to generate yield without losing ownership. Furthermore, the LNFT standard can incorporate a micro-DAO governance structure, enabling the original contributors of a Domain-Library to collectively manage its lifecycle, including updates, licensing terms, and the distribution of rewards, thereby fostering a vibrant economy around high-quality, structured knowledge.

3.2 Data-Feed

A **Data-Feed** is a user-centric, on-chain entity that enables Self-Sovereign Identity (SSI). It functions as a secure container and computational proxy for a user’s personal data. Through localized computation and cryptographic techniques, it allows users to have the effectiveness of their implicit knowledge and behavioral insights quantified without exposing raw data, and to contribute them securely to AI model training on the AXON network.

A Data-Feed, DF_j , is mathematically defined as a sextuple:

$$DF_j = \{I_j, D_{j,\text{enc}}, H_j, C_{j,\text{auth}}, S_{j,\text{usage}}, U_j\} \quad (5)$$

Where:

- DF_j : Represents the j-th user’s Data-Feed.
- I_j : An anonymous yet verifiable unique user identifier generated using zero-knowledge proofs.
- $D_{j,\text{enc}}$: Locally stored AI interaction data protected with techniques such as homomorphic encryption.
- H_j : A semantic hash of the user’s behavioral patterns, serving as a privacy-preserving representation of their personal insights.
- $C_{j,\text{auth}}$: The user-defined authorization policies, managed via Intent Signatures.
- $S_{j,\text{usage}}$: Encrypted and aggregated usage statistics.
- U_j : The model updates (e.g., gradients) generated from training on local data, which are prepared for contribution to the network.

3.2.1 Privacy as a Precondition for Contribution

In the \$AXON economic framework, robust privacy protection is not an obstacle to AI training but rather its core catalyst. This mechanism is designed to fundamentally address systemic bottlenecks in decentralized AI networks, based on the underlying logic that privacy preservation is a necessary precondition for building large-scale, high-quality collective intelligence.

This symbiotic relationship is manifested on two levels:

1. **Driving Participation at Scale:** By providing verifiable, user-controlled privacy technologies, the Data-Feed framework addresses the trust deficit inherent in traditional data economies. This trust is the cornerstone for attracting a massive user base to contribute data. Therefore, guaranteeing privacy-preserving computation is a strategic investment in expanding the scope and diversity of the network’s data sources.
2. **Enhancing Contribution Quality:** In an environment where privacy is assured, users are more more inclined to exhibit genuine, natural behavioral patterns, thereby generating high-fidelity ‘effectiveness signals’. This significantly improves the average quality and signal fidelity of data fed into AI models and effectively reduces “noise data” resulting from user dissimulation or inauthentic behavior.

It is crucial to clarify that AXON’s privacy framework employs a layered design to resolve the potential conflict between privacy and contribution validation. Raw data never leaves the user’s local environment. The network exclusively receives and processes model updates (e.g., gradients) that have been treated with formal privacy-enhancing technologies. Specifically, we employ an (ϵ, δ) -differential privacy mechanism, where the privacy budget, ϵ , can be configured by the user (e.g., defaulting to a value of 1.0) to ensure individual contributions cannot be reverse-engineered from aggregated results. The contribution validation process, therefore, operates solely on these privacy-preserving updates, never on the raw data. A detailed exposition of the privacy-enhancing technologies will be provided in AXON Chain Documentation.

3.2.2 Effectiveness Quantification via FAP

The AXON network moves beyond simplistic metrics to measure user contributions. The effectiveness of a user’s contribution, encapsulated in the model update U_j , is determined exclusively by the Federated Attribution Proof (FAP) mechanism, as detailed in Chapter 4. FAP holistically assesses the contribution’s marginal impact on improving the global model’s performance (e.g., accuracy, loss reduction). This process generates a definitive Effectiveness Index for each contribution, ensuring that rewards are strictly aligned with real, verifiable improvements to the network’s collective intelligence. This approach sidesteps the need for separate, potentially redundant quality heuristics, making FAP the single source of truth for contribution assessment.

3.2.3 User Sovereignty through Intent Signatures

To give users unprecedented control over their data, the Data-Feed introduces the concept of “Intent Signatures.” Instead of relying on opaque and complex privacy policies, a user grants data access by cryptographically signing a simple, human-readable, and machine-verifiable statement of intent. For example, an intent could be: “I, user I_j , authorize the use of my interaction data from t_1 to t_2 for the sole purpose of training ‘AI Recommender Model X’, under a differential privacy budget of $\epsilon = 0.5$. This authorization is revocable.”

These signed intents are recorded on-chain and serve as immutable, auditable proof of consent. Compute-Grids must verify a valid Intent Signature before processing any data from a Data-Feed. This mechanism transforms data sharing from a binary “opt-in/opt-out” choice into a granular, dynamic, and fully user-sovereign process, forming the bedrock of trust for the Data-Feed ecosystem.

3.3 Compute-Grid

Definition 3.3: A **Compute-Grid** is the distributed computing infrastructure of the AXON network, serving as the standardized unit for executing AI model training, validation, and inference tasks. It constitutes the execution layer of the network’s “Proof-of-Learning” paradigm, transforming traditional consensus costs into a process of creating intelligence with tangible economic utility.

The Compute-Grid is central to \$AXON’s implementation of the “Proof of Useful Work” philosophy. Unlike PoW mechanisms that waste vast energy on undifferentiated hashing, every computation cycle of a Compute-Grid is directed toward executing AI tasks that benefit the ecosystem. This design not only fundamentally enhances the network’s capital and energy efficiency but also achieves an inherent alignment between its security and development objectives.

A Compute-Grid, CG_k , is mathematically defined as a septuple:

$$CG_k = \{I_k, P_k, H_k, R_k, L_k, V_k, E_k\} \quad (6)$$

Where:

- CG_k : Represents the k-th Compute-Grid.
- I_k : A unique identifier derived from the node’s public key or decentralized identity (DID).
- P_k : The node’s heterogeneous performance profile, $\{P_{cpu}, P_{gpu}, P_{tpu}\}$, quantified in standard units like TFLOPS to enable optimal task scheduling.
- H_k : The node’s verified track record of historical task completion, serving as proof of its reliability.
- R_k : A dynamic reliability score calculated from historical success rates, uptime, and validation results, serving as a key factor in task allocation and reward weighting.
- L_k : The node’s encrypted geographical location information, used to optimize network latency and meet data sovereignty requirements.
- V_k : A sequence of cryptographic proofs (e.g., ZK-SNARKs or TEE remote attestations) generated by the node to attest to the honest completion of computational tasks.
- E_k : The node’s energy efficiency ratio (e.g., TFLOPS/Watt), a core metric used by the network to incentivize green, sustainable computation.

To precisely and fairly quantify the comprehensive contribution of a Compute-Grid, we designed a formula, expressed as an integral, to calculate its total computational contribution, $C_{k,comp}$. This formula ensures rewards are fully aligned with a node’s performance, reliability, energy efficiency, and effective task completion:

$$C_{k,comp} = \int_0^T \left[\left(\sum_{t \in \{cpu, gpu, tpu\}} P_{k,t}(s) \cdot \eta_t \right) \cdot R_k(s) \cdot E_k(s) \cdot T_k(s) \right] ds \quad (7)$$

The instantaneous contribution within the integral is a product of four key components:

- Standardized Effective Computation ($\sum P_{k,t}(s) \cdot \eta_t$): Aggregates the various types of raw computational power ($P_{k,t}$) provided by the node at time s , weighted by task-type coefficients (η_t), to yield a standardized measure of compute capability.
- Reliability Weighting ($R_k(s)$): The node’s dynamic reliability score acts as a multiplier, ensuring that stable and trustworthy nodes receive greater rewards while penalizing instability or malicious behavior.
- Energy Efficiency Incentive ($E_k(s)$): The node’s energy efficiency ratio serves as a reward factor, creating an economic incentive for the network to prioritize greener computation and promote ecological sustainability.

- Task Validity Factor ($T_k(s)$): A weighted value that ensures computation is counted as a contribution only when performing valid, network-verified AI tasks, thus preventing resource abuse.

Through this integral formula, the AXON network achieves a comprehensive, dynamic, and effectiveness-driven measurement of computational contribution, ensuring that every unit of invested computing resource serves the growth of the network’s collective intelligence.

4 Core Algorithm: Federated Attribution Proof (FAP)

Federated Attribution Proof (FAP) is the core algorithmic engine that enables AXON Network to achieve precise effectiveness distribution and trustworthy incentives. It is a comprehensive framework designed to quantify, in a cryptographically secure and mathematically rigorous manner, the specific effectiveness of each participant’s contribution (e.g., data or computation) to every AI service output. By solving the fundamental challenge of “contribution measurement in decentralized collaboration”, FAP transforms abstract collaborative actions into verifiable, auditable, on-chain records of effectiveness.

4.1 Algorithm Overview

FAP is not a singular algorithm but a closed-loop system comprising four core modules. It deeply integrates the privacy-preserving features of federated learning with a sophisticated effectiveness quantification mechanism, ensuring the entire attribution and reward process is transparent, fair, and efficient.

Definition 4.1: Federated Attribution Proof, F , is a quadruple consisting of four functions: Global Aggregation, Local Training, Attribution Calculation, and Reward Distribution.

$$\mathcal{F} = \{L, G, A, R\} \quad (8)$$

Where:

- L is the **Local Training** function: Executes on a user’s Data-Feed or a Compute-Grid associated with a Domain-Library, leveraging local data to train a model and generate updates.
- G is the **Global Model Aggregation** function: Aggregates local model updates from numerous participants into a next-generation global model in a secure and efficient manner.
- A is the **Attribution Calculation** function: The core of FAP, which, after an AI model produces an output, retrospectively calculates the Effectiveness Index (E) of each contribution.
- R is the **Reward Distribution** function: A protocol-level function that utilizes the outputs from the Attribution Calculation function (A) to inform the final distribution of AXON rewards, as detailed in the network’s economic model (Chapter 6).

Together, these four modules form a complete “train-aggregate-apply-attribute-reward” loop, ensuring that every instance of effectiveness generation within the network can be accurately traced back to its source contributors.

4.2 The FAP Loop in Detail

4.2.1 Local Training (L) and Global Aggregation (G)

The FAP process begins with the standard federated learning steps. The Local Training function (L) is executed on the participant’s device (e.g., a Data-Feed), where local data is used to compute a model update (e.g., a gradient vector) without exposing the raw data. Subsequently, the Global Aggregation function (G) securely collects these updates from multiple participants and aggregates them to produce an improved global model. These two functions provide the necessary inputs for the most critical part of FAP: Attribution Calculation.

4.2.2 Attribution Calculation (A): Quantifying Effectiveness

The Attribution Calculation function (A) is the technical heart of FAP. It employs distinct methodologies to compute the Effectiveness Index (E) for contributions from Data-Feeds and Domain-Libraries, reflecting their different characteristics.

4.2.2.1 Gradient-based Attribution for Data-Feeds

For contributions from Data-Feeds, which typically consist of model updates (U_j) from real-time interactions, FAP calculates the Effectiveness Index by measuring the update’s uniqueness and magnitude relative to the collective. This serves as a direct proxy for the contribution’s impact on the global model in a given federated learning round.

However, relying solely on the gradient norm (i.e., the magnitude of the update) for attribution presents a significant vulnerability. For instance, a malicious actor could submit an update with a large magnitude that provides no benefit—or is even detrimental—to the model’s performance in order to fraudulently claim rewards. To address this vulnerability, we have designed a multi-dimensional gradient attribution model that holistically assesses an update based on its magnitude, quality, and directional consistency.

The Effectiveness Index for a Data-Feed(DF_j), $E_{DF,j}^{(t)}$, is calculated as a weighted sum of three scores:

$$E_{DF,j}^{(t)} = \omega_m \cdot S_{mag}^{(t)}(j) + \omega_q \cdot S_{qual}^{(t)}(j) + \omega_d \cdot S_{dir}^{(t)}(j) \quad (9)$$

Where $\omega_m, \omega_q, \omega_d$ are weighting coefficients set by AXON DAO governance, satisfying $\omega_m + \omega_q + \omega_d = 1$. The three score components are defined as follows:

1. **Magnitude Score (S_{mag}):** This component represents the incentive to reward contributions that propose a significant update to the global model.

$$S_{mag}^{(t)}(j) = \frac{\|U_j^{(t)} - M_{glob}^{(t-1)}\|_2}{\sum_k \|U_k^{(t)} - M_{glob}^{(t-1)}\|_2} \quad (10)$$

2. **Quality Score (S_{qual}):** This component directly measures the real-world impact of a gradient update by evaluating its performance on a standardized, confidential validation set held by the network.

$$S_{qual}^{(t)}(j) = \max \left(0, \frac{L_{val}(M_{glob}^{(t-1)}) - L_{val}(M_{glob}^{(t-1)} + \eta U_j^{(t)})}{L_{val}(M_{glob}^{(t-1)})} \right) \quad (11)$$

- Where $L_{val}(\cdot)$ is the loss function evaluated on the standard validation set, which is maintained and updated by the network via the DVT process (detailed in Chapter 5).

- η is a standardized learning rate parameter.
 - This score quantifies the proportional reduction in loss on the validation set. A harmful update that increases the loss will receive a quality score of 0.
3. **Directional Consistency Score (Score_{dir}):** This component rewards contributions that align with the collective optimization direction of the network’s most trusted participants, thereby penalizing outliers and adversarial updates.

$$S_{dir}^{(t)}(j) = \frac{\langle U_j^{(t)}, \bar{U}_{trusted}^{(t)} \rangle}{\|U_j^{(t)}\| \cdot \|\bar{U}_{trusted}^{(t)}\|} \quad (12)$$

- Where $\bar{U}_{trusted}^{(t)} = \frac{1}{|S|} \sum k \in SU_k^{(t)}$ is the average gradient update from a set S of high-reputation nodes in the current round.
- Membership in the trusted set S is determined by a node’s multi-dimensional reputation score (Sec. 5.3.3.2) and veAXON holdings (Sec. 6.5).
- This score, representing the cosine similarity between the two vectors, measures alignment. It effectively identifies contributions that are consistent with the collective intelligence.

This multi-dimensional evaluation mechanism ensures that:

- Large-magnitude but low-quality updates are penalized with a low quality score;
- Updates that deviate from the global optimization path are penalized with a low directional score;
- High-quality, fine-tuning contributions are fairly rewarded for their quality and consistency, even if their magnitude is small.

4.2.2.2 Semantic Fingerprint Attribution for Domain-Libraries

For large-scale, static datasets within Domain-Libraries, direct gradient-based attribution is computationally prohibitive. To overcome this, the A function uses an innovative method based on semantic fingerprints. This approach shifts focus from low-level parameter changes to the semantic similarity between source data and model outputs.

However, a naive implementation of semantic attribution is vulnerable to manipulation, such as "rarity gaming," where contributors inject rare but irrelevant features to inflate scores. To counteract this, the AXON network implements a robust, multi-faceted attribution formula that incorporates a similarity threshold, a smoothed inverse-frequency weight based on differentially private counts, and a task-relevance feedback loop.

The attribution score of a Domain-Library DL_i for a given output o_t is calculated as follows:

$$E_{DL,i}(o_t) = \sum_{s \in o_t} \left[\max_{d \in DL_i} \mathbb{1}_{\text{Sim} > \tau} \cdot \text{Sim}(F_s(s), F_s(d)) \right] \cdot \frac{|s|}{|o_t|} \cdot w_{idf}(d) \cdot Q_{task}(DL_i, o_t) \quad (13)$$

The components of this formula are:

- $\mathbb{1}_{\text{Sim} > \tau}$: An indicator function that applies a minimum similarity threshold, τ . Matches with a similarity score below this domain-specific threshold are discarded, filtering out noise and irrelevant connections.
- $w_{idf}(d)$: A smoothed and capped inverse document frequency (IDF) weight that prevents gaming.

$$w_{idf}(d) = \min \left(\text{cap}, \log \left(\frac{N_{corp} + \mu}{f_{DP}(d) + \mu} \right) \right) \quad (14)$$

- $f_{DPP}(d)$ is the global frequency of a data feature d across all Domain-Libraries, estimated using a network-wide differential privacy mechanism (e.g., with parameters $\epsilon = 0.1, \delta = 10^{-6}$). This allows for global rarity assessment without compromising the privacy of individual libraries.
- N_{corp} is the total number of documents in the network, μ is a smoothing parameter, and cap is an upper bound (e.g., 3.0) to prevent extreme weights for artificially rare features.
- $Q_{task}(DL_i, o_t)$: A task-relevance quality factor that creates a crucial feedback loop. It measures the marginal performance improvement that the Domain-Library DL_i contributes to the specific AI task that produced the output o_t .

$$Q_{task}(DL_i, o_t) = \max(0, \text{Perf}(M_{base} \cup DL_i) - \text{Perf}(M_{base})) \quad (15)$$

Where $\text{Perf}(\cdot)$ is the performance metric (e.g., accuracy, F1-score) for the task, and M_{base} represents the model trained without the specific contribution of DL_i . This ensures that even a highly similar contribution receives zero reward if it fails to improve the model’s actual performance.

4.2.2.3 Phased Evolution of Semantic Fingerprints

To resolve the circular dependency of requiring advanced AI models to bootstrap an AI network, the AXON network employs a phased evolution strategy for its semantic fingerprints.

Phase 1: Bootstrap (Network Launch)

In the initial phase, fingerprints are generated using statistical, model-free methods that do not rely on pre-trained embeddings.

$$F_{s,i}^{(1)} = H_l(H_s(d_i) \oplus H_m(d_i) \oplus \text{TF-IDF}(d_i)) \quad (16)$$

Where $H_s(\cdot)$ and $H_m(\cdot)$ are structural and set-based hashes, and $\text{TF-IDF}(d_i)$ is the traditional term frequency-inverse document frequency vector.

Phase 2: Growth (Network Matures)

Once the network has accumulated sufficient data to train its own foundational models, it introduces lightweight semantic embeddings, blending them with the statistical fingerprint for stability.

$$F_{s,i}^{(2)} = H_l(w_{sem} \cdot \text{Emb}_{light}(d_i) \oplus (1 - w_{sem}) \cdot F_{s,i}^{(1)}) \quad (17)$$

Where Emb_{light} is a lightweight embedding model trained exclusively on data within the AXON ecosystem, and w_{sem} is a governance-set weight.

Phase 3: Maturity (Full Semantic Power)

In its mature state, the network transitions to using powerful, state-of-the-art embedding models for maximum attribution accuracy, confident that these models are a product of the network’s own collective intelligence.

$$F_{s,i}^{(3)} = H_l(\text{Emb}(d_i) \oplus H_s(d_i) \oplus H_m(d_i)) \quad (18)$$

4.2.3 Reward Distribution (R): Linking Effectiveness to Incentives

The Reward Distribution function (R) is not an algorithm executed by a single node, but rather a protocol-level process that connects the outputs of the Attribution Calculation (A) to the network’s incentive mech-

anism. The effectiveness scores ($E_{p,j}$ and $E_{k,i}$) calculated by the A function are aggregated and recorded on-chain within the block's Attribution Root. These immutable scores serve as the authoritative input for the Proof-of-Learning (PoL) reward distribution detailed in Chapter 6. Specifically, they determine the pro-rata allocation of the 60% of block rewards designated for data contributors, ensuring that AXON token incentives are distributed in direct proportion to the verifiable effectiveness of each participant's contribution.

4.3 Comprehensive Contribution Aggregation

To create a long-term, holistic reputation for each network participant i , FAP aggregates their effectiveness scores across different dimensions (data, compute) using a time-decay weighting.

4.3.1 Time-Decay Factor

To ensure the contribution metric is time-sensitive, we introduce an exponential time-decay factor D , which assigns greater weight to more recent contributions:

$$D(t, \tau) = e^{-\lambda_d \cdot \frac{t-\tau}{\tau_0}} \quad (19)$$

Where:

- $D(t, \tau)$: The decay weight applied at the current time t to a contribution made at a past time τ .
- λ_d : The decay constant.
- τ_0 : A time-scale parameter that defines the "half-life" of a contribution's value.

4.3.2 Comprehensive Contribution Score

A participant i 's comprehensive contribution score up to time t , denoted $C_{i,total}^{(t)}$, is the time-decayed sum of all their historical contributions. This creates a dynamic ledger of reputation and long-term value:

$$C_{i,total}^{(t)} = \sum_{\tau=1}^t D(t, \tau) \cdot [\omega_d \cdot E_{d,i}^{(\tau)} + \omega_c \cdot C_{c,i}^{(\tau)}] \quad (20)$$

Where:

- $C_{i,total}^{(t)}$: The comprehensive contribution score of participant i at time t .
- $E_{d,i}^{(\tau)}$: The total data effectiveness contribution (from either a Data-Feed or Domain-Library) made at time τ , as calculated by the attribution algorithm A (Sec. 4.2.2).
- $C_{c,i}^{(\tau)}$: The compute contribution made at time τ , calculated by the formula $C_{k,comp}$ (Sec. 3.3).
- ω_d, ω_c : Global weighting coefficients for data and compute contributions, respectively, set via governance by the AXON DAO.

Through this comprehensive contribution model, FAP unifies all effectiveness-generating activities across the network into a single fair, transparent, and dynamic measurement framework. This provides a robust algorithmic foundation for the stable operation and virtuous incentive structure of the \$AXON economic ecosystem.

5 AXON Network Block Generation Mechanism

The consensus layer of the AXON network is a direct embodiment of its “**Proof of Useful Work**” philosophy. To replace traditional consensus protocols that involve significant redundancy in energy and computational resources, we have designed an innovative **Data-Driven Block Generation (DDBG)** mechanism. This mechanism directly links the right to produce blocks with real, quantifiable effectiveness contributions to the network’s ecosystem, thereby transforming the cost of maintaining network security into an investment that drives the growth of the network’s collective intelligence.

5.1 Mechanism Overview

The core principle of DDBG is that block generation should not be determined by undifferentiated computational power, but should instead be led by the contributors who provide the most effectiveness to the network. Here, effectiveness is precisely measured by the Federated Attribution Proof (FAP) defined in Chapter 4. This mechanism adheres to four core principles to ensure the network’s security, efficiency, and fairness.

1. **Principle of Data Effectiveness Primacy:** A node’s block production weight is positively correlated with the effectiveness of the data it contributes, which is continuously evaluated by the FAP algorithm. This ensures that contributors of the network’s core resource—high-effectiveness data—are maximally incentivized.
2. **Principle of Contribution Immediacy:** A node’s recent contributions are given greater weight in the block producer candidacy assessment. This provides immediate feedback to active participants and ensures the network can respond effectively to the most effective current activities.
3. **Principle of Privacy by Design:** The entire block production process, including data validation and attribution, is conducted within a cryptographic and privacy-preserving framework, consistent with the security models of the Data-Feed and Domain-Library defined in Chapter 3.
4. **Principle of Contribution Diversity:** By holistically evaluating data, computational, and stake-based contributions, DDBG effectively prevents resource monopolies in any single dimension (such as a computational monopoly) and encourages a more resilient, healthy ecosystem composed of diverse contributors.

5.2 Block Proposer Selection Process

Each block production cycle is a structured, multi-stage process designed to select the network’s most significant contributor during that cycle to propose the new block.

5.2.1 Candidate Node Selection

At the start of each block production cycle t , the system selects a pool of candidate block producers based on each node’s comprehensive contribution score, S_{comp} , within the latest evaluation window, T_w . This score is a weighted composite of data, compute, and economic stake contributions:

$$S_{comp,i}^{(t)} = w_d \cdot S_{data,i}^{(t)} + w_c \cdot S_{compute,i}^{(t)} + w_s \cdot S_{stake,i}^{(t)} \quad (21)$$

Here, w_d, w_c, w_s are weighting coefficients determined by AXON DAO governance (e.g., $w_d = 0.5, w_c = 0.3, w_s = 0.2$), a configuration that reflects the network's strategic focus on data contribution as the primary effectiveness driver. The individual scores are calculated as follows:

- **Data Contribution Score** ($S_{data,i}^{(t)}$): This score quantifies the total value of node i 's data contributions within the evaluation window T_w . It is calculated by taking the time-decayed sum of the node's data effectiveness scores (as defined in Chapter 4) and normalizing it across the network:

$$S_{data,i}^{(t)} = \frac{\sum_{\tau=t-T_w}^t D(t, \tau) \cdot E_{d,i}^{(\tau)}}{\sum_{j \in N} \sum_{\tau=t-T_w}^t D(t, \tau) \cdot E_{d,j}^{(\tau)}} \quad (22)$$

Where $E_{d,i}^{(\tau)}$ is the data effectiveness score of node i at time τ (from FAP), $D(t, \tau)$ is the time-decay factor, and N is the set of all network nodes. This formula ensures the score reflects both cumulative contribution and the heightened importance of recent activity.

- **Compute Contribution Score** ($S_{compute,i}^{(t)}$): Similarly, this score quantifies the effectiveness of node i 's computational contributions during the window, based on its compute contribution $C_{k,comp}$ (defined in Sec. 3.3) and normalized:

$$S_{compute,i}^{(t)} = \frac{\sum_{\tau=t-T_w}^t D(t, \tau) \cdot C_{c,i}^{(\tau)}}{\sum_{j \in N} \sum_{\tau=t-T_w}^t D(t, \tau) \cdot C_{c,j}^{(\tau)}} \quad (23)$$

Where $C_{c,i}^{(\tau)}$ is the compute contribution of node i at time τ .

- **Staking Score** ($S_{stake,i}^{(t)}$): This score measures a node's economic commitment. To balance the influence of capital against long-term commitment, the formula is designed as follows:

$$S_{stake,i}^{(t)} = T_{lock,i} \cdot \log(1 + V_i) \quad (24)$$

Where V_i is the amount of \$AXON tokens staked by node i , and $T_{lock,i}$ is its lock-up period. The logarithmic function is applied to the staked amount, V_i , to significantly dampen the outsized advantage of raw capital, creating diminishing returns as the stake grows. Concurrently, the lock-up period, $T_{lock,i}$, acts as a linear multiplier, strongly rewarding nodes for making long-term commitments to network security.

The system ranks all nodes by their $S_{comp,i}^{(t)}$ score, and the top- K nodes qualify to form the candidate pool for the current block production round.

5.2.2 Data Validation Task (DVT) Allocation

Nodes in the candidate pool are assigned one or more **Data Validation Tasks (DVT)**. DVTs represent concrete "useful work," requiring nodes to use their resources to validate new network contributions. Key task types include:

1. **Domain-Library Quality Validation:** Cross-validating newly submitted Domain-Libraries on dimensions like accuracy and completeness, according to the $E_{i,prior}$ metrics from Sec. 3.1.

2. **Federated Attribution Proof Calculation:** Executing the FAP algorithm (Chapter 4) for recent AI training tasks to compute the Effectiveness Index of each data source.
3. **Attribution Proof Verification:** Cryptographically verifying a sample of attribution proofs (e.g., ZK-SNARKs) generated by other nodes to ensure the integrity of the FAP results, reinforcing the "complex execution vs. efficient verification" security model.

Task allocation is not random but probabilistic, weighted by the candidacy score $S_{c,i}^{(t)}$, ensuring that higher-contribution nodes undertake more critical validation responsibilities:

$$P(\text{assign}_{i,j}) \propto S_{c,i}^{(t)} \quad (25)$$

Where $P(\text{assign}_{i,j})$ is the probability of assigning task j to node i .

It is important to clarify how a single compute provider's resources are allocated between general AI computation and network validation. The AXON network employs a time-division multiplexing model. During non-block-producing periods, a node's resources are dedicated to executing AI training tasks, for which it earns standard compute contribution rewards. When selected as a block proposer candidate, a fraction of its resources are temporarily allocated to performing its assigned DVTs. These validation tasks are designed to be computationally bounded, typically consuming only 5-10% of a node's operational capacity for that cycle. This design ensures high resource utilization and avoids idle capacity, fully realizing the "Proof of Useful Work" principle by integrating security tasks seamlessly with value-generative AI computation.

5.2.3 Final Proposer Selection

Candidate nodes compete for the final block proposal right by completing DVTs with high efficiency and accuracy. This process involves three steps:

Step 1: Submit Proof-of-Effectiveness Node i submits its Proof-of-Effectiveness ($P_{e,i}$) for the current cycle, which is a cryptographic digest of its latest contributions.

Step 2: Execute DVT and Generate Validation Report Node i executes its assigned DVTs and generates a validation report, $R_{v,i}$, containing the results.

Step 3: Calculate Final Proposer Score The system calculates a final score, $S_{f,i}$, based on the quality of the node's contribution proof, the accuracy of its DVT completion, and its speed.

To mitigate centralization pressures that favor nodes with the lowest latency, the time-based factor's influence is curtailed. The final score is a weighted average, with the vast majority of the weight given to the quality and accuracy of the work performed, rather than pure speed.

The final proposer score, $S_{f,i}$, is calculated as:

$$S_{f,i} = 0.6 \cdot S_{poe,i} + 0.3 \cdot S_{vr,i} + 0.1 \cdot F_{t,i} \quad (26)$$

Where $S_{poe,i}$ is the score for the Proof-of-Effectiveness, $S_{vr,i}$ is the score for the validation report's accuracy, and the time-based reward factor, $F_{t,i}$, is calculated using a piecewise function:

$$F_{t,i} = \begin{cases} 1.0 & \text{if } t_i \leq t_{min} + t_{SLA} \\ \max\left(F_{min}, e^{-\lambda_t \cdot \frac{t_i - (t_{min} + t_{SLA})}{t_{avg}}}\right) & \text{if } t_{min} + t_{SLA} < t_i \leq t_{max} \\ F_{min} & \text{if } t_i > t_{max} \end{cases} \quad (27)$$

The components of this function are:

- t_{SLA} : A Service Level Agreement window (e.g., 2 seconds), within which all nodes who submit their results receive the maximum time factor of 1.0. This rewards a wide range of timely participants, not just the single fastest.
- t_{max} : A cutoff time (e.g., $t_{min} + 3 \cdot t_{avg}$) after which submissions are still accepted but receive the minimum time score.
- F_{min} : A minimum time factor (e.g., 0.5) to ensure that slower but correct nodes still receive a baseline reward for their work.
- λ_t : A gentle decay rate (e.g., 0.5) for submissions that arrive after the SLA window but before the cutoff.

This structure ensures that while timeliness is encouraged, it does not dominate the selection process, fostering a more decentralized and fair environment for block proposers.

5.3 Consensus Architecture and System Properties

5.3.1 Block Structure and Consensus Finality

The structure of an AXON block is defined by three core Merkle Trees, whose root hashes are recorded in the block header to ensure data integrity and immutability.

1. **World State Tree**: This is a global, continuously evolving tree that records the state of all entities on the network, including account balances, LNFT (Domain-Library) metadata, Compute-Grid reputation scores, and veAXON lock-up information. The `worldStateRoot` in the block header represents the latest world state snapshot after executing all transactions in the current block.
2. **Transaction Tree**: This tree contains all transactions packaged within the current block. The `transactionsRoot` in the block header ensures that the set of transactions within the block is itself immutable.
3. **Receipts Tree**: This tree contains the outcome of each transaction's execution, such as transaction fee and success status. The `receiptsRoot` provides a verifiable proof for the execution results of transactions.

The Block Body is simpler in contrast, primarily containing the detailed list of all transactions within the block.

A simplified pseudo-code representation, sufficient to illustrate the core architecture, is as follows:

```
//Conceptual structure of an AXON Block
```

```
struct Block {
```

```

    header: BlockHeader,
    body: BlockBody,
}

struct BlockHeader {
    // — Standard Fields —
    block_height: u64,
    timestamp: u64,
    previous_hash: Hash,

    // — Core State Tree Roots —
    world_state_root: Hash, // Root of the global world state
    transactions_root: Hash, // Root of the block's transaction tree
    receipts_root: Hash, // Root of the block's transaction receipts tree

    // — PoL Consensus Proof Fields —
    attribution_root: Hash, // Root of FAP attribution results
    effectiveness_root: Hash, // Root of DVT validation results

    // ... other fields, e.g., proposer address
}

struct BlockBody {
    // Contains a list of all transactions for the block
    // (e.g., transfers, staking, governance votes, LNFT operations)
    transactions: Vec<Transaction>,
}

```

To achieve rapid finality, AXON employs a hybrid consensus mechanism: DDBG efficiently selects the block proposer, after which a validation committee of high-stake nodes performs BFT voting. A block receiving signatures from over 2/3 of the committee is confirmed as final. In the unlikely event of a fork, the network resolves it using the Heaviest Effectiveness Chain rule, selecting the fork that has generated the most verifiable effectiveness for the ecosystem.

5.3.2 Incentives and Penalties

Rewards from each new block are precisely distributed across the value creation chain: 60% to data contributors based on FAP results, 30% to the block proposer, and 10% to the validation committee. To deter malicious activity, the network enforces a strict penalty mechanism. Any node proven to have acted maliciously (e.g., submitting false validation reports, manipulating attribution results) will have its stake slashed. The slashed amount, S_{slash} , is scaled according to the severity of the offense:

$$S_{\text{slash}} = \min(S_{\text{total}} \cdot F_{\text{sev}}, S_{\text{total}}) \quad (28)$$

Where S_{total} is the node's total stake, and F_{sev} is a severity factor for specific malicious acts, as defined by the AXON DAO.

5.3.3 Performance and Security Guarantees

The AXON network ensures its security and resilience against various attacks through a multi-layered design. Its core security philosophy is derived from an innovative application of the asymmetry principle, deeply integrating network security with valuable computation.

5.3.3.1 Core Anti-Attack Mechanism Based on Asymmetry

We adapt the core principle of asymmetry from Bitcoin’s Proof-of-Work (PoW) consensus, elevating it from arbitrary hash calculations to “useful work” that benefits the network. This asymmetry is manifested as follows:

- **Bitcoin’s Asymmetry:** Finding a valid solution (mining) is computationally difficult, requiring immense trial-and-error, whereas verifying a given solution is trivial, requiring only a single hash calculation.
- **AXON’s Asymmetry:** Executing “useful work” (e.g., a DVT) is computationally complex, while verifying the result of that work is efficient and inexpensive.

This design makes it computationally infeasible for attackers to act maliciously by simply forging results. For example, when a block proposer executes a complex FAP algorithm, they must generate a verifiable computation proof (e.g., a ZK-SNARK). Forging such a proof is computationally infeasible, as its difficulty is equivalent to that of breaking the underlying cryptographic primitives. The BFT validation committee can then efficiently verify this proof without re-executing the entire complex task. This “complex execution vs. efficient verification” asymmetry forms the core of AXON’s consensus security.

5.3.3.2 Multi-Dimensional Identity Reputation System

To further mitigate Sybil attacks, AXON implements a multi-dimensional identity reputation system, R_{sybil} . A node’s reputation is not solely dependent on its stake but is a composite score derived from multiple factors:

$$R_{\text{sybil}} = f(S_{\text{stake}}, H_{\text{data}}, P_{\text{compute}}, T_{\text{active}}) \quad (29)$$

Where:

- S_{stake} : The node’s economic stake and lock-up duration.
- H_{data} : The node’s historical record of data contributions (effectiveness and quantity).
- P_{compute} : The node’s historical record of computational contributions.
- T_{active} : The node’s sustained uptime and stability.

A new node, even one with substantial capital, must earn a high reputation score over time through consistent and effective contributions. This design significantly increases the cost and timeframe required to mount a successful Sybil attack.

5.3.3.3 Major Attack Vectors and Defenses

The AXON network architecture is designed with specific defenses against several major attack vectors:

1. **Data Poisoning Attacks:** This attack involves submitting malicious data intended to degrade a model’s performance. AXON defends against this through a multi-layered validation process:

- DVT cross-validation subjects newly submitted Domain-Libraries to scrutiny by other network participants.
 - The FAP algorithm’s Quality Score (S_{qual}) and Directional Consistency Score (S_{dir}) will assign a near-zero effectiveness score to contributions that harm model performance or deviate from the consensus optimization path.
 - The long-term reputation system will automatically flag and penalize participants who repeatedly submit low-quality or harmful contributions.
2. **51% Attack Resistance:** In the context of AXON’s Proof-of-Learning, a 51% attack is substantially more difficult than in traditional networks. An attacker would need to simultaneously control a majority of three distinct resource classes:
- 51% of the network’s FAP-verified data contribution effectiveness.
 - 51% of the network’s compute resources to dominate the DVT process.
 - 51% of the staked tokens to influence the staking score.

The immense cost and complexity of acquiring a majority in all three dimensions make a classic 51% attack economically and practically infeasible.

3. **Long-Term Sybil Attack Mitigation:** Even if an attacker manages to create and maintain a set of Sybil nodes over a long period, their impact is quantitatively limited by the system’s ongoing behavioral analysis. The total impact of a Sybil cluster can be modeled as:

$$\text{Attack}_{\text{impact}} \leq \frac{n_{\text{sybil}} \cdot \bar{C}_{\text{sybil}}}{N \cdot \bar{C}_{\text{honest}}} \cdot (1 - \rho) \quad (30)$$

Where ρ represents the system’s Sybil detection rate (e.g., $\rho > 0.7$), achieved through behavioral pattern analysis and other heuristics. This ensures that even undetected Sybil clusters have a constrained and diminishing influence on the network.

A comprehensive security model and its formal verification will be provided in a dedicated blockchain technical whitepaper.

5.3.4 Performance and Dynamic Adjustment

While prioritizing security and decentralization, the AXON network is engineered as a high-performance blockchain capable of supporting large-scale AI applications and a vibrant on-chain economy. Its performance characteristics are designed to rival modern systems.

- **High Throughput and Low Latency:** The network is designed with an adaptive block time, benchmarked at 3 seconds but dynamically adjustable between 2-5 seconds based on network load and the computational complexity of DVTs. This design strikes a critical balance between rapid transaction confirmation and the need for sufficient time to execute meaningful validation work. At a 3-second block time, combined with parallel transaction processing and an optimized execution environment, AXON can achieve a theoretical throughput of approximately 5,000-10,000 transactions per second (TPS), providing a robust foundation for large-scale AI applications and real-time data contributions.
- **Rapid Finality:** Leveraging the BFT consensus performed by the validation committee after each DDBG-led block proposal, transactions achieve finality within seconds. This rapid confirmation provides the certainty required for high-value operations and seamless integration with other services, eliminating the uncertainty of probabilistic finality found in older consensus models.

- **Scalable Architecture:** The “Proof of Useful Work” model inherently scales network utility with security. As more computational resources join the network to increase its security, they are simultaneously directed towards executing more AI training and validation tasks (DVTs). This creates a positive feedback loop where network growth directly translates into increased capacity for the AI ecosystem, ensuring that performance scales alongside adoption.

6 The \$AXON Tokenomic Model

The \$AXON tokenomic model is the value accrual layer for its core “Proof-of-Learning” philosophy. It is designed to foster a self-regulating, incentive-compatible, and value-driven decentralized AI economy. Unlike traditional models that rely on a single dimension such as compute power or capital, \$AXON’s economic system employs a sophisticated, multi-dimensional reward mechanism. It directly links token issuance to all forms of “useful work”—including the computation that secures the network, the contribution of high-value knowledge, and privacy-preserving data interactions—ensuring that every newly issued \$AXON token is anchored to the tangible growth of the network’s collective intelligence.

6.1 Foundational Parameters and Initial Allocation

To ensure a predictable, long-term deflationary trajectory, the \$AXON token economy is established with the following immutable genesis parameters:

- **Total Supply (Hard Cap):** 86,000,000,000 \$AXON. This hard cap, inspired by the approximate number of neurons in the human brain, guarantees the token’s eventual scarcity and anchors the project’s vision in building a collective intelligence.
- **Initial Allocation**
 - **Testnet & Early Contributor Rewards (5%):** 4,300,000,000 \$AXON. This portion will be distributed as a one-time allocation to individuals and teams who made outstanding contributions to network testing, code audits, community building, and the nascent ecosystem prior to the mainnet launch. This allocation rewards early supporters and ensures the network launches with a cohort of experienced, engaged participants.
 - **Fair Launch Pool (95%):** 81,700,000,000 \$AXON. The remaining 95% of the total supply is allocated to a fair launch pool. These tokens can only be emitted via the reward mechanisms defined in this chapter, with no pre-allocation for the team or investors.
- **Minimum Emission Rate (λ):** 0.01.
- **Decay Coefficient (a):** 2.5.

6.2 Dynamic Emission Model

To ensure transparency and predictability in token issuance, \$AXON utilizes a deterministic, decaying emission model tied to the circulating supply. This model provides high initial incentives to bootstrap the network rapidly, while naturally transitioning toward a deflationary state over time.

The total emission rate $v(x)$ during any time period Δt is given by:

$$v(x) = \lambda + (1 - \lambda) \cdot a^{-\frac{x}{N-x}} \quad (31)$$

Where:

- $v(x)$ is the token emission rate when the circulating supply is x .
- x is the cumulative total of issued AXON tokens (including the 5% initial allocation).
- N is the total supply cap of 86 billion.
- λ is the minimum emission rate.
- a is the decay coefficient.

To provide a precise definition of the issuance mechanism, we first define the token supply's dynamic equation in its continuous, theoretical form:

$$\frac{dX(t)}{dt} = v(X(t)) \cdot (N - X(t)) \quad (32)$$

This is discretized at the block level to determine the number of new tokens minted in each block:

$$\Delta X_{block} = v(X_{current}) \cdot (N - X_{current}) \cdot \frac{T_{block}}{T_{year}} \quad (33)$$

Where $X_{current}$ is the circulating supply at the time of the block's creation, T_{block} is the target block time (e.g., 3 seconds), and T_{year} is the number of seconds in a year (31,536,000).

Where $X(t)$ is the cumulative issuance at time t , and $v(X)$ is the instantaneous annualized emission rate.

6.3 Proof-of-Learning (PoL) Reward Distribution

The \$AXON reward mechanism is not separate from block production; rather, it is the intrinsic logic governing the fine-grained distribution of the total block reward. All newly emitted tokens from each block are strictly allocated to PoL contributors across different dimensions according to the **60% / 30% / 10%** structure defined in Chapter 5.

6.3.1 Compute & Security Rewards (40% of Block Rewards)

This reward component, constituting 40% of total block rewards, incentivizes the core computational work required to secure the network, execute transactions, and validate blocks. It is allocated as follows:

- **Proposer Reward (30%):** 30% of all tokens emitted in a block are granted directly to that block's proposer.
- **Validator Rewards (10%):** 10% of all tokens emitted in a block are distributed among the members of that block's validation committee.

This component is critical for attracting professional nodes and enterprise-level participants to commit hardware resources, thereby establishing the network's foundational security.

6.3.2 Data and Knowledge Contribution Rewards (60% of Total Rewards)

This component, constituting the majority 60% of block rewards, is the core expression of the "Proof-of-Learning" philosophy. It directly rewards the contribution of high-effectiveness data, which is the lifeblood of the AI ecosystem. To facilitate a strategic shift from initially seeding the network with high-quality knowledge to later encouraging broad-based personal data participation, this 60% reward pool is subject to a dynamic weighting between Domain-Library contributions and Data-Feed contributions.

6.3.2.1 Dynamic Weighting Mechanism

The total data contribution reward, R_{data} , constitutes 60% of the total block reward. This pool is then partitioned according to the dynamic weights $W_k(t)$ and $W_p(t)$:

- **Layer 2: Domain-Library Contributions**

- **Weight**

$$W_{DL}(t) = 0.15 + 0.3 \cdot e^{-\frac{t}{t_k}} \quad (34)$$

- **Design Rationale:** The reward weight for Domain-Libraries begins at 0.45 (representing 75% of the total data rewards) and, over time t , decays exponentially toward a long-term steady state of 0.15 (25% of data rewards). This is designed to heavily incentivize the early integration of large-scale, high-quality, structured knowledge bases, which provide a robust foundation for the network's AI models.

- **Layer 3: Data-Feed Contributions**

- **Weight**

$$W_{DF}(t) = 0.6 - W_{DL}(t) \quad (35)$$

- **Design Rationale:** Conversely, the reward weight for Data-Feeds grows from an initial 0.15 (25% of data rewards) to a long-term weight of 0.45 (75% of data rewards). As the network matures, this design shifts the incentive focus toward the broadest base of participants, encouraging the contribution of diverse, personalized, long-tail data. This enhances the generalization capabilities and the degree of decentralization of the network's AI models.

The respective reward pools are then calculated as:

- $R_{pool,DL}(t) = \frac{W_{DL}(t)}{0.6} \cdot R_{data}$
- $R_{pool,DF}(t) = \frac{W_{DF}(t)}{0.6} \cdot R_{data}$

6.3.2.2 Reward Allocation Process

Once the reward pool sizes for each contribution type are determined, the specific allocation of rewards adheres strictly to the outputs of the Federated Attribution Proof (FAP). The reward for a contributor i is calculated in three steps:

Step 1: Calculate Base Attribution Contribution (B_i)

This step uses the FAP algorithm (Chapter 4) to calculate a baseline effectiveness score for each participant. This score reflects the fundamental effectiveness their data contributed to the outputs of all AI tasks on the network.

$$B_i = \sum_o I_o \cdot A_m(o, i) \quad (36)$$

- $A_m(o, i)$: The attribution score from FAP for contributor i 's role in AI task output o .
- I_o : The Task Impact Weight of the AI task output o . This is a dimensionless multiplier set by the AXON DAO through on-chain governance. It serves as a strategic tool for the community to incentivize data contributions towards AI tasks deemed more critical for the ecosystem's development at a given time. For example, the DAO could vote to assign a higher I_o to tasks related to training a new foundational model versus fine-tuning an existing one.

Step 2: Calculate Effective Contribution Score ($C_{eff,i}$)

Two multipliers are applied to the base contribution to derive the final score used for reward allocation.

$$C_{eff,i} = B_i \cdot Q_i \cdot S_i \quad (37)$$

- **Q_i : Quality Factor.** This is a post-attribution multiplier derived from data metadata (e.g., accuracy, timeliness, integrity) as evaluated by the DVT process (Chapter 5). It serves as a distinct mechanism to reward well-formatted and verifiable data, ensuring it is orthogonal to the FAP's core effectiveness calculation.
- **S_i : Staking Boost Factor.** As the core of the veAXON mechanism, this factor translates a user's economic commitment directly into a reward advantage. S_i is a monotonically increasing function of a user's veAXON balance ($V_{ve,i}$). Its formula is:

$$S_i = 1 + \gamma \cdot \log(1 + V_{ve,i}) \quad (38)$$

Where γ is a governance-adjustable parameter.

Step 3: Calculate Final Reward

A contributor's final reward is their pro-rata share of their category's total effective contribution score, applied to that category's reward pool.

- Reward for Domain-Library Contributor i :

$$R_{DL,i} = R_{pool,DL}(t) \cdot \frac{C_{eff,DL,i}}{\sum_j C_{eff,DL,j}} \quad (39)$$

- Reward for Data-Feed Contributor i :

$$R_{DF,i}(t) = R_{pool,DF}(t) \cdot \frac{C_{eff,DF,i}}{\sum_j C_{eff,DF,j}} \quad (40)$$

6.4 Token Utility and Value Capture

The intrinsic value of the \$AXON token is underpinned by its core utilities within the ecosystem. These functions collectively drive demand for the token, enabling it to capture value from the network's growth.

6.4.1 Network Security and Consensus

- **Validator Staking:** To become a validator node on the AXON network, participants must stake a specified amount of \$AXON tokens. This requirement is governed by a Dynamic Staking Threshold to

ensure the cost of securing the network scales with its economic value. The threshold, which defines the minimum stake required for entry, is calculated as:

$$S_{sec}(t) = S_{base} \cdot \left(1 + \alpha \cdot \frac{N_{val}(t)}{N_{tgt}} + \beta \cdot \frac{V_{tvl}(t)}{V_{tgt}} \right) \quad (41)$$

Where:

- S_{base} : The Base Staking Threshold.
 - $N_{val}(t)$: The current number of validators.
 - N_{tgt} : The target number of validators.
 - $V_{tvl}(t)$: The network’s Total Value Locked (TVL).
 - V_{tgt} : A target TVL milestone.
 - α, β : Adjustment coefficients set by governance.
- **Data Quality Bond:** Creating a Domain-Library requires staking \$AXON as a “quality bond,” which serves as an economic guarantee for the authenticity and quality of the provided data. If the data is found to be malicious or of severely low quality by on-chain governance, this bond can be slashed.

6.4.2 Economic Incentives and Governance

- **Contribution Reward Boost:** As detailed in Section 6.3.2, locking \$AXON to receive veAXON directly increases a user’s share of data and knowledge contribution rewards via the Staking Boost Factor (S_i).
- **Governance Participation:** Locking \$AXON tokens to obtain the governance token, veAXON, grants users direct participation in shaping the network’s future, as detailed in Section 6.5.

6.4.3 Transaction Fees and DAO Treasury

The AXON network operates without a traditional, fluctuating gas fee market. Instead, it employs a predictable, fixed-fee model for transactions. These fees are not burned but are distributed to network participants, creating a closed-loop economy. All transaction fees are split dynamically between the block proposer who includes the transaction and the community-governed DAO treasury.

- **Fee Rate Table:** Transaction fees are calculated based on a Base Fee Unit (e.g., defined by governance as 0.001 \$AXON) multiplied by a rate specific to the transaction type. This ensures predictability and governance flexibility.

Transaction Type	Fee Rate	Rationale
Simple Transfer	1x	Base transaction cost.
Staking Operation	10x	Involves more complex state changes.
LNFT Mint/Update	100x	High-value operation, requires storage updates.
Governance Vote	5x	Encourages participation while preventing spam.
Compute-Grid Registration	500x	Onboarding a core infrastructure provider.
More	DAO Defined	For future extensibility.

Table 1: Fee Rate Table

- **Dynamic Fee Distribution:** The split of fees between the block proposer and the DAO is not static. It is governed by a formula that reduces the DAO's share as its treasury approaches a pre-defined cap, preventing excessive centralization of funds.

– The DAO's share (P_{DAO}) of a given transaction fee is calculated as:

$$P_{DAO} = \begin{cases} 0.5 \cdot \left(1 - \frac{T_{current}}{T_{cap}}\right)^2 & \text{if } T_{current} < T_{cap} \\ 0 & \text{if } T_{current} \geq T_{cap} \end{cases} \quad (42)$$

– The proposer's share is then $P_{proposer} = 1 - P_{DAO}$.

Where:

- $T_{current}$ is the current balance of the DAO treasury.
- T_{cap} is the DAO treasury's hard cap, set by governance (e.g., 1% of total supply, or 860,000,000 \$AXON).

This mechanism ensures the DAO is well-funded in its early stages (receiving up to 50% of all fees) to foster ecosystem growth, while its accumulation of power is automatically curtailed as it matures.

- **DAO Treasury Function:** The DAO treasury, funded by these fees, is controlled by veAXON holders. They vote on the allocation of these funds for ecosystem grants, public goods, developer incentives, and strategic initiatives, enabling community-directed reinvestment of the protocol's value.

6.5 veAXON: The Core of Governance and Incentives

veAXON is the governance cornerstone and incentive multiplier of the \$AXON ecosystem, designed to reward long-term commitment and align protocol control with long-term value contributors.

6.5.1 veAXON Calculation Mechanism

Users lock \$AXON tokens ($V_{st,i}$) for a chosen duration ($T_{lock,i}$) to receive veAXON. The veAXON balance, $V_{ve,i}$, is calculated using a formula that applies a logarithmic function to the staked amount to mitigate the Matthew effect, while rewarding longer lock-up periods.

$$V_{ve,i} = k \cdot \log(1 + V_{st,i}) \cdot \left(1 + \frac{T_{lock,i}}{T_{max}}\right) \quad (43)$$

Where:

- k is a coefficient used to adjust the value of veAXON to an appropriate range.
- $V_{st,i}$ is the amount of \$AXON tokens staked by user i . The logarithmic function significantly dampens the outsized advantage of raw capital, ensuring that governance power does not concentrate excessively in the hands of a few large holders.
- $T_{lock,i}$ is the lock-up duration chosen by the user.
- T_{max} is the maximum possible lock-up duration (e.g., 4 years), set by governance. The linear multiplier for lock-up duration strongly incentivizes long-term commitment to the network.

6.5.2 Governance Weight Calculation

To further promote decentralization, a user’s final governance power, $P_{gov,i}$, is not directly proportional to their veAXON balance but is calculated using a concave function to diminish the voting power of single large holders:

$$P_{gov,i} = \frac{(V_{ve,i})^{0.75}}{\sum_j (V_{ve,j})^{0.75}} \quad (44)$$

6.5.3 Core Rights of veAXON

Holding veAXON is the exclusive key to participating in AXON’s deep value distribution. Its core rights include:

1. **Protocol Governance Power:** veAXON holders have ultimate control over the protocol, including voting on key parameters and the allocation of DAO treasury funds.
2. **Contribution Reward Boost:** A user’s veAXON balance directly calculates the Staking Boost Factor (S_i), significantly increasing their share of data contribution rewards.
3. **Ecosystem Airdrop Weight:** veAXON holders will receive greater weight or priority eligibility for future airdrops from new projects launching within the AXON ecosystem.

6.6 Token Allocation Projections and Strategic Guidance

This section translates the preceding models into a tangible forecast of the network’s evolution, based on the 86 billion token supply and the 60/30/10 reward structure.

6.6.1 Participant Personas

- **Professional Compute Providers (Pro Miners):** These are entities or individuals with specialized, large-scale computational hardware (e.g., GPU farms). Their primary motivation is return on investment (ROI). They contribute their hardware as productive assets to the AXON network, primarily earning block rewards by running validator nodes and competing to propose blocks. They form the bedrock of the network’s initial security and performance.
- **Enterprise Users:** These are typically companies or institutions with existing business needs. They possess two key resources: in-house compute clusters for AI/ML development and high-quality, proprietary, structured datasets from their business domains. They engage with AXON for two main reasons: first, to connect idle compute resources to the network to earn rewards and offset hardware costs; and second, to securely monetize their proprietary data by creating Domain-Libraries to earn data contribution rewards.
- **Retail Participants:** This represents the largest and most diverse group, including crypto enthusiasts, technologists, and general users. Their resources are typically more limited, ranging from a mid-to-high-end personal computer to simply their time and attention. They participate through various low-barrier methods, such as running light nodes, staking small amounts of AXON, and most importantly, contributing personal data and behavioral feedback via Data-Feeds.

- **AI Enthusiasts:** This group forms the core consumer base of the network’s application layer and is a primary source of value creation. Their main driver is access to the novel and powerful AI applications and agents deployed on AXON. Their key resource is their interaction with these AIs. In AXON’s economic model, every meaningful interaction (e.g., a query, an annotation, or feedback) can be captured via Data-Feeds, validated as a contribution, and rewarded with tokens. They are essential in transforming the network from mere infrastructure into a vibrant ecosystem.

6.6.2 Illustrative Allocation and Emission Projections

This section combines the token emission model with behavioral assumptions for each user type to provide a clear, data-driven roadmap of the token allocation’s projected evolution.

6.6.2.1 Token Emission Rate Projections

First, using the dynamic emission model from Section 6.2, we project the token emissions per block at different stages of the network’s lifecycle.

- **Core Parameters & Assumptions**

- **Total Supply (N):** 86,000,000,000 \$AXON
- **Minimum Emission Rate (λ):** 0.01
- **Decay Coefficient (a):** 2.5
- **Block Time:** 3 seconds

- **Emission Projections by Stage**

- **Early Stage (10% of Supply Issued - 8.6B \$AXON)**
 - * With $x/(N - x) \approx 0.111$, the annualized emission rate is $v(x) \approx 90.4\%$.
 - * Projected emissions per block: $0.904 \cdot (86B - 8.6B) \cdot \frac{3}{31,536,000} \approx 6,658$ \$AXON.
 - * Rationale: An aggressive initial emission rate is designed to rapidly attract foundational participants.
- **Mid-Stage (50% of Supply Issued - 43B \$AXON)**
 - * With $x/(N - x) = 1$, the annualized emission rate is $v(x) = 40.6\%$.
 - * Projected emissions per block: $\approx 1,446$ \$AXON.
 - * Rationale: Emissions have decreased significantly but remain attractive, fostering stable growth.
- **Late Stage (90% of Supply Issued - 77.4B \$AXON)**
 - * With $x/(N - x) = 9$, the annualized emission rate is $v(x) \approx 1.025\%$.
 - * Projected emissions per block: ≈ 8.12 \$AXON.
 - * Rationale: Emissions have substantially declined. The network enters maturity, where its value is primarily sustained by internal economic activity (fees).

6.6.2.2 Detailed Allocation Analysis

The following table illustrates the projected distribution of token rewards among participant groups across the three stages, incorporating the 60/40 reward split and the dynamic data reward weighting.

- **Early Stage (Outbreak Period)**

Reward Pool (Share of Emissions)	Early Stage ($t \rightarrow 0$)	Mid-Stage ($t \approx t_k$)	Late Stage ($t \rightarrow \infty$)
Compute & Security (40%)			
Professional Miners	70% ($\sim 28\%$)	50% ($\sim 20\%$)	30% ($\sim 12\%$)
Enterprise Users	30% ($\sim 12\%$)	45% ($\sim 18\%$)	60% ($\sim 24\%$)
Retail Participants	0%	5% ($\sim 2\%$)	10% ($\sim 4\%$)
Data Contribution (60%)			
Domain-Library Rewards (45% $\rightarrow$ 22.5% $\rightarrow$ 15%)			
Enterprise Users	90% ($\sim 40.5\%$)	70% ($\sim 15.75\%$)	50% ($\sim 7.5\%$)
Retail/Enthusiasts	10% ($\sim 4.5\%$)	30% ($\sim 6.75\%$)	50% ($\sim 7.5\%$)
Data-Feed Rewards (15% $\rightarrow$ 37.5% $\rightarrow$ 45%)			
Retail/Enthusiasts	100% ($\sim 15\%$)	100% ($\sim 37.5\%$)	100% ($\sim 45\%$)

Table 2: Distribution of the Reward Pool Across Stages (Professional Style)

- **Strategic Goal:** To rapidly establish foundational compute power and high-quality knowledge bases.
- **Allocation Landscape:** Enterprise Users (12% compute + 40.5% data = 52.5%) and Professional Compute Providers (28%) are projected to earn over 80% of all emitted tokens. High block rewards and heavily weighted Domain-Library rewards create powerful incentives for foundational contributions. The Retail & AI Enthusiast cohort (4.5% Domain-Library + 15% Data-Feed = 19.5%) starts with a significant base, driven by the active personal data reward mechanism.
- **Mid-Stage (Growth Period)**
 - **Strategic Goal:** To maintain network stability while expanding to a broader user base and more diverse data sources.
 - **Allocation Landscape:** The share for Professional Compute Providers declines to 20%. Enterprise Users (18% compute + 15.75% data = 33.75%) remain a backbone. The most significant shift is the rise of the Retail & AI Enthusiast share to a commanding 46.25% (2% compute + 6.75% Domain-Library + 37.5% Data-Feed). This is driven by the increased weight for personal data rewards, making everyday user interactions economically valuable and fueling rapid community expansion.
- **Late Stage (Maturity Period)**
 - **Strategic Goal:** To achieve a high degree of decentralization, foster the continuous contribution of long-tail data, and support a thriving application ecosystem.
 - **Allocation Landscape:** By this stage, the share for Retail & AI Enthusiasts is projected to reach 56.5% (4% compute + 7.5% Domain-Library + 45% Data-Feed), establishing them as the primary contributing force. Enterprise Users’ share stabilizes around 31.5%. The dynamic weighting mechanism will have successfully steered the network’s incentive structure from favoring early capital-intensive contributions to rewarding broad community engagement and long-tail data, fostering a highly decentralized distribution of network ownership.

6.7 Game Theoretic Analysis and Incentive Compatibility

To ensure the robustness of the \$AXON economic model, we analyze the system's incentive compatibility from a game-theoretic perspective. Our goal is to demonstrate that for rational participants, honest and effective contribution is the optimal strategy.

6.7.1 Basic Game Setup

We model the system as a repeated game $G = (N, S, U)$, where:

- $N = 1, 2, \dots, n$ is the set of participants.
- S_i is the strategy space for participant i , which includes Honest Contribution (s_h), Malicious Manipulation (s_m), and Free-riding (s_f).
- $U_i : S \rightarrow \mathbb{R}$ is the utility function for participant i .

6.7.2 Incentive Compatibility of Data Contribution

Theorem 1: Under the FAP mechanism, honestly contributing high-quality data is a weakly dominant strategy.

Proof: Let the utility function for a data contributor i be defined as:

$$U_i(s_i, s_{-i}) = R_i(s_i, s_{-i}) - C_i(s_i) - P_i(s_i) \quad (45)$$

Where R_i is the reward calculated by the FAP and PoL mechanisms, C_i is the cost of contribution (e.g., data acquisition, processing), and P_i is the potential penalty from slashing.

We analyze the expected utility for each strategy:

1. **Honest Contribution (s_h):** The participant submits genuine, high-quality data. The utility is

$$U_i(s_h) = R_i(s_h) - C_i(s_h) \quad (46)$$

The system is calibrated such that for valid contributions, the expected reward exceeds the cost, making $U_i(s_h) > 0$.

2. **Malicious Manipulation (s_m):** The participant attempts to maximize rewards by submitting low-quality or synthetic data designed to game the attribution algorithm. The expected utility is

$$U_i(s_m) = \mathbb{P}(\text{undetected}) \cdot R_i(s_m) - C_i(s_m) - \mathbb{P}(\text{detected}) \cdot S_{slash} \quad (47)$$

Given the multi-dimensional FAP validation (Sec. 4.2.2) and DVT cross-checks (Sec. 5.2.2), the probability of detection, $\mathbb{P}(\text{detected})$, is high. Since the slashing penalty S_{slash} is designed to be significantly larger than any potential illicit gain, the expected utility becomes negative: $E[U_i(s_m)] < 0$.

3. **Free-riding (s_f):** The participant does not contribute, incurring no cost but also receiving no reward. The utility is

$$U_i(s_f) = 0 \quad (48)$$

From this analysis, we establish the preference relation: $U_i(s_h) > U_i(s_f) > E[U_i(s_m)]$. Therefore, honest contribution (s_h) is the weakly dominant strategy.

6.7.3 Nash Equilibrium in Compute Contribution

Theorem 2: In the DDBG mechanism, a stable mixed-strategy Nash Equilibrium exists, where high-performance and high-commitment nodes consistently contribute computation, ensuring network security and functionality.

Proof Outline: The decision for a node to provide computation is based on its expected profit. A node i 's optimal response depends on its comprehensive contribution score $S_{c,i}^{(t)}$, which determines its probability of being selected as a block proposer. We can define a participation threshold, θ , where contributing is profitable if the expected reward exceeds the operational cost ($C_{compute}$).

$$\theta = \frac{C_{compute}}{E[R_{block}] \cdot P(\text{proposer} | S_{c,i}^{(t)})} \quad (49)$$

- If a node's score $S_{c,i}^{(t)} > \theta$, its optimal strategy is to continue contributing.
- If $S_{c,i}^{(t)} < \theta$, the node is better off exiting.

This dynamic creates a stable equilibrium. Nodes with superior hardware, higher data contribution scores, and larger veAXON stakes will consistently find it profitable to participate, forming the reliable core of the network's compute layer.

6.7.4 Collusion Resistance Analysis

Theorem 3: Under the veAXON governance model, the cost of a large-scale collusion attack to seize control of the network is designed to be prohibitively greater than its maximum potential benefit.

Proof Outline: Consider a coalition $M \subset N$ of m malicious nodes attempting to control governance. The cost of this attack is:

$$\text{Cost}_{\text{collusion}} = \sum_{i \in M} V_{st,i} + C_{\text{coordinate}} + \text{Risk}_{\text{slash}} \quad (50)$$

This includes the total value of staked tokens ($V_{st,i}$), which is at risk, plus significant coordination costs ($C_{\text{coordinate}}$).

The maximum benefit is the ability to control the allocation of protocol rewards and treasury funds. However, the coalition's governance power, $P_{gov,M}$, is subject to the concave function defined in Formula:

$$P_{gov,M} = \frac{(\sum_{i \in M} V_{ve,i})^{0.75}}{\sum_{j \in N} (V_{ve,j})^{0.75}} \quad (51)$$

Due to the fractional exponent (0.75), the governance power of the coalition increases at a sub-linear rate relative to the linear increase in their staked capital cost. This diminishing return on investment makes it economically irrational to attempt to acquire a controlling share of governance power, as the cost quadratically outpaces the marginal gain in voting power, thus securing the network against such economic attacks.

7 Conclusion

AXON Network addresses the critical challenges of centralization in the modern AI landscape—data monopolies, computational barriers, and inequitable value distribution. This paper has detailed a comprehensive tokenomic framework, centered on a single native asset, the \$AXON token, to foster a decentralized, contribution-oriented, and sustainable AI value network.

The cornerstone of this framework is the PoL philosophy, which transforms network security from a resource-consuming cost into a value-creative process. This is achieved through a synergetic architecture where the DDBG consensus mechanism allocates block production rights based on verifiable contributions. The FAP algorithm serves as the system’s core engine, precisely and transparently quantifying the effectiveness of all data and knowledge contributions to AI model training. This ensures that rewards are distributed equitably, in direct proportion to the value each participant provides to the network’s collective intelligence.

Furthermore, the economic model incorporates robust mechanisms for long-term sustainability and decentralized governance. A deterministic, decaying emission schedule guarantees eventual scarcity, while a novel, dynamic fee distribution model funds a community-governed DAO without risking fund centralization. The veAXON governance system elegantly balances the influence of capital with long-term commitment, aligning participants’ incentives with the network’s enduring success. Together, these components provide a viable and robust economic foundation for large-scale, decentralized AI collaboration.

Future research will focus on extending the FAP framework to cross-chain environments, developing next-generation privacy-preserving technologies, and refining the dynamic economic parameters through algorithmic governance. As the ecosystem matures, the AXON tokenomic model will continue to evolve, laying a solid foundation for an open, inclusive, and efficient intelligent future, built and owned by its contributors.

Appendix

A Glossary of Terms and Notation

Symbol	Description	First Defined
A.1 Core Entities		
DL_i	A Domain-Library, representing a knowledge vault.	Eq. (1)
DF_j	A Data-Feed, representing a user’s data pod.	Eq. (5)
CG_k	A Compute-Grid, representing a compute node.	Eq. (6)
C_i	The list of contributors for Domain-Library DL_i .	Eq. (1)
A.2 Effectiveness and Attribution		
E_i	Comprehensive Effectiveness Index of a Domain-Library.	Eq. (2)
$E_{i,prior}$	A priori (static) Effectiveness Index.	Eq. (3)
$E_{i,post}$	A posteriori (dynamic) Effectiveness Index.	Eq. (4)
$E_{i,peer}$	Peer-review score for a Domain-Library’s effectiveness.	Eq. (2)
$E_{DF,j}^{(t)}$	Effectiveness Index for a Data-Feed j at time t .	Eq. (9)
$E_{DL,i}(o_t)$	Attribution score of Domain-Library DL_i for output o_t .	Eq. (13)
A.3 Economic Model		
N	Total token supply cap (86 billion).	Sec. 6.1
$v(x)$	Annualized token emission rate function.	Eq. (31)

Symbol	Description	First Defined
λ	Minimum emission rate (0.01).	Sec. 6.1
a	Decay coefficient for emission rate (2.5).	Sec. 6.1
$V_{st,i}$	Amount of \$AXON tokens staked by user i .	Eq. (43)
$V_{ve,i}$	veAXON (governance token) balance of user i .	Eq. (43)
$P_{gov,i}$	Final governance power of user i .	Eq. (44)
$W_{DL}(t)$	Dynamic reward weight for Domain-Library contributions.	Eq. (34)
$W_{DF}(t)$	Dynamic reward weight for Data-Feed contributions.	Eq. (35)

A.4 Consensus and Block Production

$S_{comp,i}^{(t)}$	Comprehensive contribution score of a node for candidacy.	Eq. (21)
$S_{data,i}^{(t)}$	Data contribution component of the comprehensive score.	Eq. (22)
$S_{compute,i}^{(t)}$	Compute contribution component of the comprehensive score.	Eq. (23)
$S_{stake,i}^{(t)}$	Staking component of the comprehensive score.	Eq. (24)
$S_{f,i}$	Final proposer score for a candidate node.	Eq. (26)
$F_{t,i}$	Time-based reward factor for DVT submission.	Eq. (27)

A.5 Time and Indexing Conventions

Symbol	Description
t, τ	Time indices, often representing current and past time, respectively.
i, j, k	General indices for participants, often specialized for Domain-Libraries (DL_i), Data-Feeds (DF_j), and Compute-Grids (CG_k).
s, d	Indices for semantic segments (s) and data items (d).
o_t	An AI task output generated at time t .

A.6 Mathematical Operations

Symbol	Description
$\oplus$	Concatenation or fusion operation for creating fingerprints.
$\langle \cdot, \cdot \rangle$	Inner product of two vectors.
$\ \cdot \ $	L2 norm (magnitude) of a vector.
$\mathbb{1}_{\text{condition}}$	Indicator function (evaluates to 1 if the condition is true, 0 otherwise).
$\mathbb{P}(\cdot)$	Probability of an event.
$\mathbb{E}[\cdot]$	Expected value of a random variable.